Oracle Architectural Components

History of Oracle:

1977:

Larry Ellison, Bob Miner, and Ed Oates found software development Laboratories (SDL).

1979:

SDL releases Oracle Version 2, its first Commercial relational database management system.

1983:

SDL becomes Oracle Corporation and releases Oracle database Version 3 with SQL Support.

1986:

Oracle goes public, trading on NASDAQ.

Late 1980s - Early 1990s:

Releases Oracle7 with PL/SQL, stored procedures and triggers.

1997:

Oracle8 is released, Supporting OOP and multimedia data types.

2008:
Oracle Database 11g released focus on performance, security and manageability.
2010:
Acquires Sun Microsystems, gaining and like java and MySQL.
2013:
Oracle database 12c introduces multitenant architecture for improved resource management.
2001 oracle 18c
2019:
Launches Oracle Autonomous Databases, utilizing machine learning and automation for self-driving capabilities. oracle 19c.

The Oracle database family:
=> Oracle Database 19c is the most recent version.
=> The family of database products includes:

Oracle Enterprise Edition:
- flagship database product aimed at large-scale implementations

Oracle standard edition:
- designed for small and medium-sized implementations.

Oracle Express Edition:
- entry-level and unsupported as a product database available at no charge

Oracle Database Features

Database application development features:

Database programming:
- SQL provides basic functions for data manipulation, transaction control, and record retrieval from the database.
- PL/SQL is a procedural language extension to SQL.
- PL/SQL can be used to build stored procedures and triggers, looping controls, conditional statements and error handling.

Database extensibility:
- Oracle multimedia provides text manipulation and additional image, audio, video and locator functions in the database.

◦ XML DB provide the native support for XML data type.

Database Connection features:

Oracle Net Services:

◦ Provides a scalable, secure and easy to-use high availability network infrastructure for Oracle environment.

Oracle Internet Directory:

◦ LDAP (Light weight Directory Access protocol) based directory service.

Oracle Connection Manager:

◦ An intermediate server that forwards connection requests to database servers or to other proxy servers.

Distributed database features

Distributed Queries and Transactions:

◦ Retrieve data from multiple databases

Heterogeneous Services:

◦ Allows non-Oracle data and services to be accessed from an oracle database through generic connectivity.

Data movement features:

Transportable features:

- move or copy tablespace from one database to another

Advanced Queuing and oracle Streams:

- Uses to asynchronously send msgs from one Oracle database to another.

Database Performance features:

Database parallelizations:

- Database tasks are implemented in parallel to speed the querying, tuning and maintenance of the database.

Data warehousing:

- Performance enhancements for business intelligence applications.

Database security features:

◦ Oracle includes basic security for managing user access through roles and and privileges.

- Virtual private database (VPD)
- Advanced Security Options (ASO)

Managing the Oracle Database

⇒ Self-tunning and self-managing features that make the database easier to manage.
- Automatic workload Repository (AWR)
- Automatic Database Diagnostic monitor
- Automatic memory management (AMM)
- Automatic Storage management (ASM)
- Oracle Enterprise manager.

Oracle Architecture

⇒ Describe the basic structure of oracle database.

⇒ The architecture Composed of:
- Physical components
- memory components
- Processes and Logical structure

Oracle database

- Oracle RDBMS provides an open, Comprehensive, integrated approach to information management.
- Oracle database reliable manages a large amount of data in a multiuser environment so that the user can concurrently access the same data.

Oracle Architecture Components

1) Oracle Server
   - Oracle Instance
   - Oracle Database
2) Physical Structure
3) Memory Structure
4) Logical structure

(1) Oracle Server

⇒ Oracle Server, often referred to as Oracle Database Server, is the core component of the oracle database architecture.

⇒ It serves as the foundation for storing, managing and processing data in oracle databases.

⇒ An Oracle Server:

- is a dbms that provides an open comprehensive, integrated approach to information management.
- Consists of an:
  - Oracle Instance
  - Oracle Database.

(*) Database:

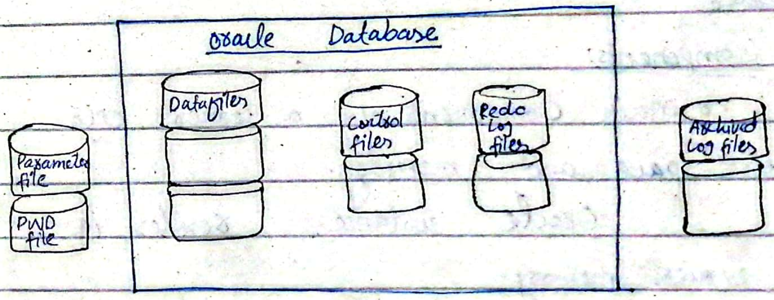
- A database is a collection of data on disk in one or more files on a database server that collects and maintains related information.
- Contains physical and logical structures such as tables, indexes, views and stored procedures that define how data is organized and access.
- Tables consist of rows and column containing related data.
- minimum requirement for a useful database is to have tables.
- Provides security mechanisms to prevent unauthorized access to data.
- Files composing a database fall into two broad categories:
  - database files
  - non-database files
- The distinction lies in what kind of data stored in each.

database files:

- Contain data and meta-data

non-database files:

- Contain initialization parameters, logging information and so forth.



(*) Oracle Instance:

=> An instance is a "collection of memory structures and background processes" that manage the oracle database.

=> it is a mean to access an oracle database.

=> Always opens one and only one database.

=> Exists in server memory, separate from the database stored on disk.

=> Composed of a large block of memory called system Global area (SGA).

=> Includes background processes that interact between SGA and database files on disk.

=> The combination of the SGA and the oracle processes is called an oracle server.

Components:

- Main Components of a server: CPUs, disk space and memory.
- Oracle instance resides in servers' memory.

Oracle real Application clusters:

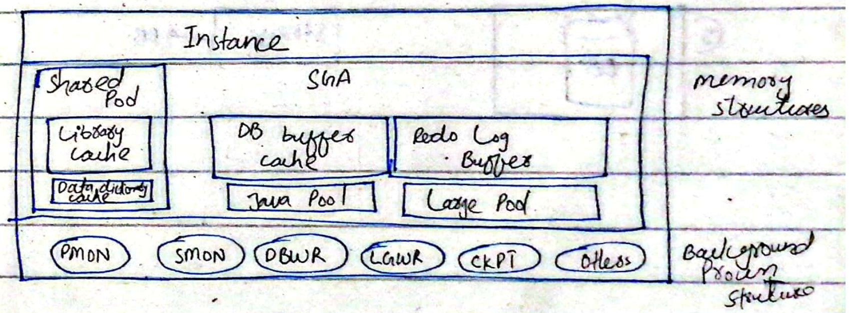
- multiple instances can share the same database.
- Instances may be in the same server or separate server connected by high speed interconnect.

Background Processes:

- Perform tasks such as memory management, I/O operations, and coordination between instances and database files.
- Essential for the functioning of oracle instances.

SGA: System global area holds data and control information shared among database users. (Shared memory)

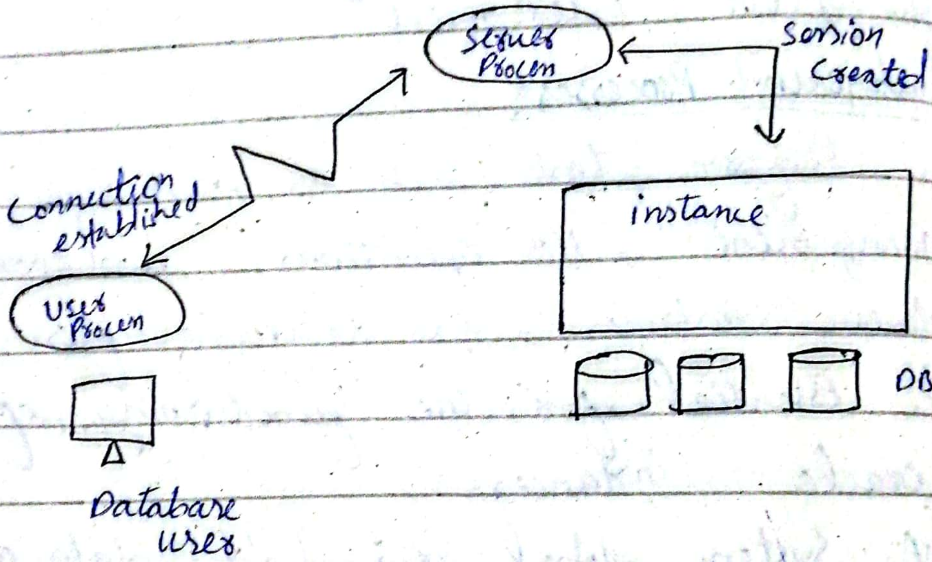
PGA: Program global area. Contains data and control information specific to a single data base session (private memory)



Establishing a Connection and Creating a Session:

⇒ Connecting to an Oracle instance:

- Establishing a user connection
- Creating a session.



for understanding: